

# Characterization of a High-Temperature DC SQUID

V-I Traces, Magnetometry, and Shapiro Step Detection

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May 2026

# Outline

Part 1: Motivation and Theoretical Foundation

Part 2: Cryogenic Experimental Setup and Noise Mitigation

Part 3: Quantitative Results and Characterization

Part 4: Shapiro Resonances and Advanced Quantum Metrology

Part 5: Conclusions and Outlook

## **Part 1: Motivation and Theoretical Foundation**

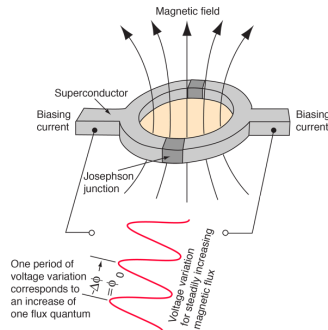
# Motivation: Frontier of High-Sensitivity Quantum Sensing

## Why Superconducting Quantum Interference?

SQUIDs exploit macroscopic quantum coherence to resolve magnetic flux below a single quantum  $\Phi_0$ , making them the ultimate flux-to-voltage transducers.

## Key Applications in Frontier Physics

- **Biomedical MEG:** Mapping brain activity [Hämäläinen 1993].
- **Scanning SQUID (SOT):** Imaging nano-spins and thermal dissipation [Vasyukov 2013, Halbertal 2016].
- **Metrology:** Phase-locked Josephson voltage standards.



SQUID loop and phase modulation schematic

# Theoretical Foundation: Josephson Effect

## Macroscopic Quantum Wave Function

Condensate is modeled by  $\Psi = \sqrt{\rho}e^{i\phi}$ . Two weak-link superconductors separated by an insulating barrier exhibit phase difference  $\delta = \phi_1 - \phi_2$  [Sólyom Vol 2, Feynman Lectures].

## Josephson Equations

- **DC Relation:** Dissipationless supercurrent

$$I = I_c \sin \delta \quad (1)$$

- **AC Relation:** Phase evolution under voltage bias  $V$

$$\dot{\delta} = \frac{2eV}{\hbar} = \frac{2\pi}{\Phi_0} V \quad (2)$$

## Josephson Non-Linear Inductance

Differentiating current yields a phase-dependent inductance:

$$L_J(\delta) = \frac{\Phi_0}{2\pi I_c \cos \delta} \quad (3)$$

This non-linear behavior makes JJs the primary building blocks of qubits, preventing harmonic over-excitation [Makk Lecture Notes].

# Theoretical Foundation: DC SQUID

## Macroscopic Interference

Two parallel JJs enclosing a loop enforce the phase constraint:

$$\delta_2 - \delta_1 = \frac{2e}{\hbar} \oint \vec{A} \cdot d\vec{r} = 2\pi \frac{\Phi_{\text{ext}}}{\Phi_0} \quad (4)$$

## Current Modulation

Supercurrent interference yields:

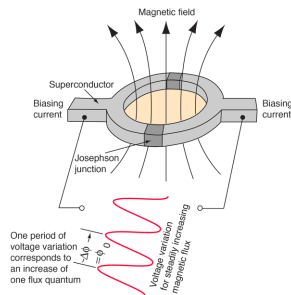
$$I_{c,\text{SQUID}} = 2I_c \left| \cos \left( \pi \frac{\Phi_{\text{ext}}}{\Phi_0} \right) \right|$$

## Voltage Transduction

Constant current bias  $I_{\text{bias}} > 2I_c$  yields a flux-dependent voltage:

$$V(\Phi) = \frac{R}{2} \sqrt{I_{\text{bias}}^2 - I_{c,\text{SQUID}}^2(\Phi)} \quad (6)$$

The SQUID translates external magnetic fields into microvolts, enabling sub-quantum magnetometry [Granata et al. 2016].



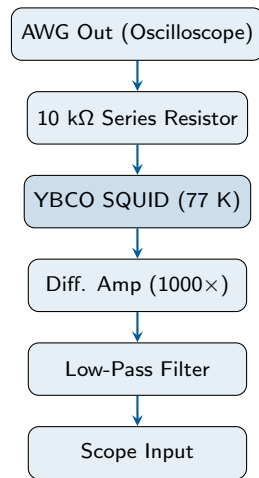
SQUID schematic and modulation

## **Part 2: Cryogenic Experimental Setup and Noise Mitigation**

# Cryogenic Experimental Setup

## High- $T_c$ SQUID Setup

- **YBCO SQUID:** grain-boundary bicrystal device cooled in liquid nitrogen ( $T = 77$  K) [SQUID.md].
- **Current Bias:** Waveform Generator driven through a  $10\text{ k}\Omega$  series resistor ( $R_{\text{series}} \gg R_N$ ).
- **Voltage Sensing:** SQUID voltage amplified  $1000\times$  by low-noise differential op-amp and filtered.
- **RF Coupling:** HP 8620C sweep oscillator delivering microwave signals (8–12 GHz) via SMA line.



Cryogenic measurement circuit schematic



# Noise Mitigation and Parametric Derivatives

## Noise Reduction

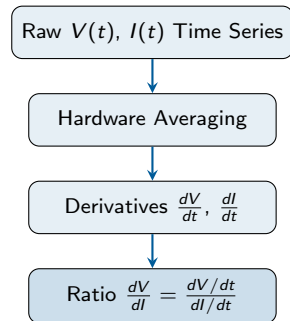
Cryogenic thermal noise requires hardware averaging ( $4\times/16\times$ ) on the oscilloscope to suppress random fluctuations.

## Parametric derivative methodology

Numerical differentiation  $dV/dI$  introduces heavy AD/DAC sample interleaving noise. We bypass this by computing time-domain parametric derivatives:

$$\frac{dV}{dI} = \frac{dV/dt}{dI/dt} \quad (7)$$

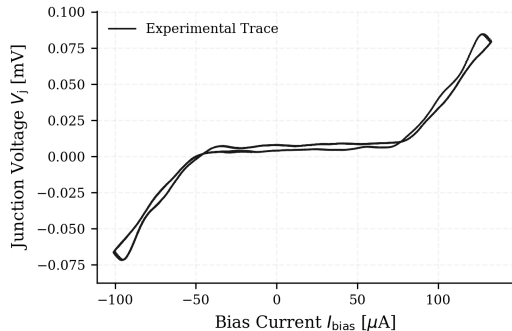
This ensures clean, physical differential resistance curves without artificial spikes [report.tex].



Parametric derivative calculation flow

## **Part 3: Quantitative Results and Characterization**

# Single Junction V-I Characteristics (Task 1)



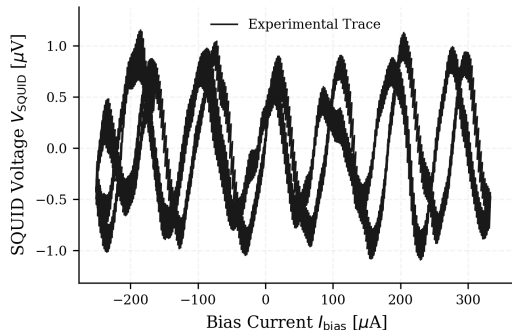
V-I curve driven at  $I_{\text{sweep}} = \pm 120 \mu\text{A}$

## Underdamped Hysteretic Transport

The single-junction sweep displays highly stable, hysteretic switching behavior.

- **Critical Current:**  $I_c = 82 \mu\text{A}$  at the lowest temperature (superconducting threshold).
- **Capacitive Hysteresis:** Hysteretic return path is characteristic of underdamped junctions where capacitance shunting is significant.
- **Gain Validation:** Junction voltage spans  $\pm 1.5 \text{ mV}$ , amplified to  $\pm 1.5 \text{ V}$  on scope, verifying differential amplifier gain of  $1000\times$ .

# SQUID Voltage Modulation under Flux Drive (Task 2)



SQUID voltage modulation under 41 Hz triangular flux drive

## Triangular Flux Sweep

We applied a triangular flux-bias current of  $\pm 300 \mu\text{A}$  at a frequency of 41 Hz through the flux line to characterise the SQUID:

- **Flux Periodicity:** Approximately 6 full flux periods (7 peaks) are clearly resolved within the sweep drive envelope.
- **Stable Cooling:** Confirms robust, stable superconducting phase locking under liquid nitrogen cooling at 77 K.
- **V-I Integration:** Demonstrates active voltage-flux transduction  $V(\Phi)$  as the basis for magnetometry.

# SQUID Voltage-Flux Transduction & Mutual Inductance (Task 3)

## Peak-to-Peak Modulation

SQUID voltage oscillates periodically under flux drive:

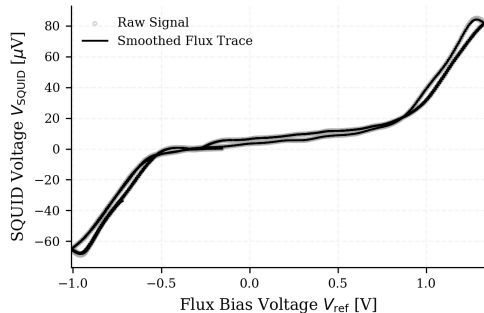
- Max peak modulation  $V_{pp} = 2.2 \mu\text{V}$  at DC biases of 1.0 V and 1.1 V.
- Suppressed to  $0.8 \mu\text{V}$  at 1.5 V bias (normal state).

## Mutual Inductance Extraction

Periodicity  $\Delta I_{\text{flux}} = 82.9 \mu\text{A}$  over 14 cycles yields:

$$M = \frac{\Phi_0}{\Delta I_{\text{flux}}} = 24.9 \text{ pH} \quad (8)$$

Manual calculation gives 25.2 pH (1.2% agreement).



Zoomed SQUID voltage vs. flux bias current

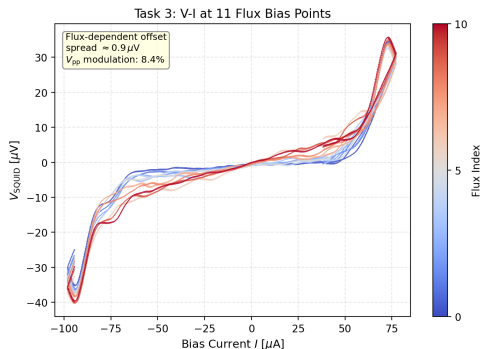
# Critical Current Modulation $I_c(\Phi)$ (Task 3b)

## Direct Observation of Quantum Interference

V-I curves recorded at 11 flux bias points demonstrate critical current modulation:

- **Voltage Offset Spread:**  $\sim 0.9 \mu\text{V}$  spread visible near zero current bias.
- **Modulation Depth:** Max modulation depth of  $6.0 \mu\text{V}$  (8.4% of peak voltage).
- **Phase Tuning:** The continuous transition confirms the macroscopic relation:

$$I_c(\Phi) = 2I_0 \left| \cos \left( \pi \frac{\Phi}{\Phi_0} \right) \right| \quad (9)$$



V-I curves at 11 flux biases spanning one period

## **Part 4: Shapiro Resonances and Advanced Quantum Metrology**

# Phase-Locking: AC Josephson Effect

## AC Josephson Phase-Locking

Under microwave irradiation at frequency  $\nu$ , the external AC field phase-locks the junction's Cooper pair transport, producing constant-voltage plateaus (Shapiro steps) in the V-I curve [Doh et al. 2005].

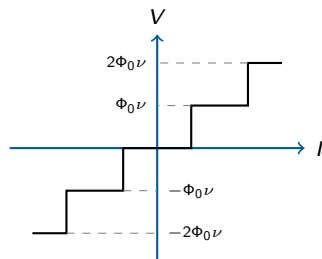
## Quantized Voltage Steps

$$\Delta V = n\Phi_0\nu = n\frac{h}{2e}\nu \quad (10)$$

For  $\nu = 10$  GHz, the expected fundamental step size is:

$$\Delta V_{\text{theory}} = 20.7 \mu\text{V} \quad (11)$$

These steps provide the modern, primary physical standard for the definition of the Volt.



Theoretical quantized Shapiro steps staircase



# Step Extraction Methodology: Gaussian LSCV-KDE

## Fragility of Ad-hoc Heuristics

Standard  $dV/dI$  derivative peak-prominences or histogram bin-widths are highly subjective. At 10 GHz, they yield step estimates ranging wildly from  $13.3 \mu\text{V}$  to  $25.6 \mu\text{V}$  depending on arbitrary tuning parameters.

## Least-Squares Cross-Validation (LSCV)

We implement an automated, parameter-free Gaussian Kernel Density Estimator (KDE). LSCV determines the optimal bandwidth  $h$  by minimizing:

$$\text{LSCV}(h) = \int \hat{f}_h^2 - \frac{2}{N} \sum \hat{f}_{-i,h}(x_i) \quad (12)$$

**Central Slicing:** We slice the central 50% current sweep region to reject normal ohmic branch noise.

## Bandwidth Comparison

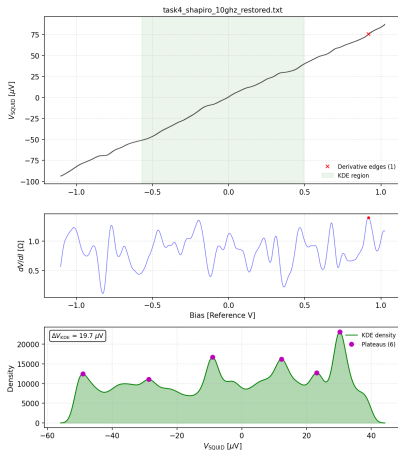
- **Silverman's rule:**  $h = 6.7 \mu\text{V}$  (oversmooths multimodal peaks, merging plateaus).
- **LSCV optimization:** selects  $h = 1.5 \mu\text{V}$ , successfully resolving discrete plateaus spacing of  $\sim 20 \mu\text{V}$  without empirical bias.

# Shapiro Step Verification at 10 GHz (Task 4)

## KDE Step Extraction

We resolve Shapiro steps using automated Gaussian KDE with LSCV bandwidth optimization:

- **Peak Resolution:** Resolves 6 symmetric peaks corresponding to step indices  $n \in \{-3, -2, -1, 0, +1, +2, +3\}$ .
- **High Accuracy:** Median spacing  $\Delta V_{\text{KDE}} = 19.7 \mu\text{V}$ , yielding only a **4.8% error** against theory ( $20.7 \mu\text{V}$ ).
- **Objective Selection:** Avoids human bias in step size extraction.



KDE step extraction workflow at 10 GHz

# Frequency Dependence and Bimodal Scattering (Task 5)

## Flux Quantum Fitting

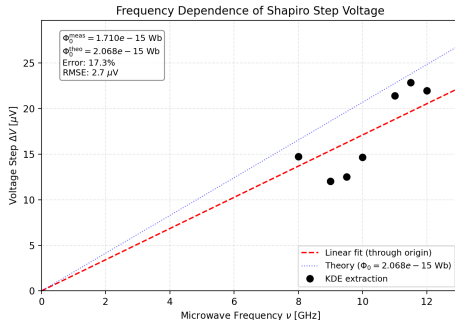
We swept frequency from 8 to 12 GHz. A linear fit through the origin ( $\Delta V = \Phi_0 \nu$ ) yields:

$$\Phi_0^{\text{meas}} = 1.71 \times 10^{-15} \text{ Wb} \quad (13)$$

This represents a 17.3% deviation from the theoretical value, with an RMSE of  $2.7 \mu\text{V}$ .

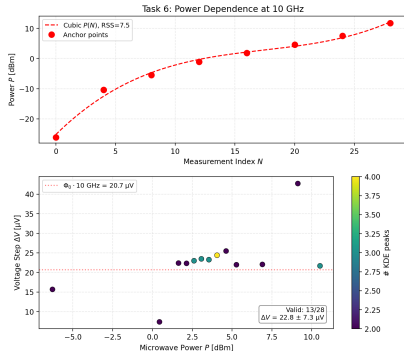
## Standing-Wave Impedance Mismatch

The data exhibits a bimodal clustering: lower frequencies (8–9.5 GHz) yield step sizes of 12–15  $\mu\text{V}$ , while higher ones (11–12 GHz) cluster at 21–23  $\mu\text{V}$ . This physical clustering is a systematic waveguide coupling effect: standing wave resonance on the open SMA line distorts power transfer, rather than a failure of the KDE extractor.



Voltage step vs. Frequency

# Power Dependence and Harmonic Steps (Task 6)



Microwave power interpolation and step size vs. power

## Power Sweep & Interpolation

Sweep markers mapped using a 3rd-order fit:

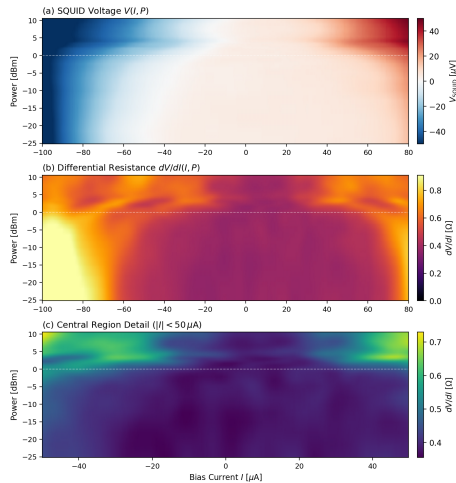
$$P(N) = a_3 N^3 + a_2 N^2 + a_1 N + a_0$$

(RSS = 7.53, parameters  $a_i$  stable).

## Onset & Second Harmonic

- **Threshold:** Steps appear above  $\sim 0$  dBm ( $N > 12$ ), averaging  $\Delta V = 23.0 \pm 7.2 \mu V$ .
- **Harmonic Step:** Resolved at high power (9.1 dBm):  $\Delta V = 41.7 \mu V \approx 2\Delta V_{\text{theory}}$ , indicating  $n = 2$  locking.

# Full Spatial Visualization of Shapiro Steps Colormaps



- **Panel (a)** shows the transition broadening of the V-I curves as power increases.
- **Panel (b)** shows the clear suppression of the critical switching current peaks.
- **Panel (c)** displays the central region detail, where the emerging structure of Shapiro steps is clearly resolved above 0 dBm.

## **Part 5: Conclusions and Outlook**

# Conclusions

## Summary of Achievements

We completed the comprehensive electrical and quantum characterization of a YBCO DC SQUID at liquid nitrogen temperature (77 K).

## Quantitative Findings:

- **Junction Transport:** Hysteretic, underdamped V-I trace yielding  $I_c = 82 \mu\text{A}$  at 77K.
- **Flux Magnetometry:** Voltage modulation depth of  $2.2 \mu\text{V}$  yielding  $M = 24.9 \text{ pH}$  (1.2% agreement to theory).
- **Quantum Metrology:** Linear voltage steps verified under microwave irradiation. KDE extraction at 10 GHz yields  $\Delta V = 19.7 \mu\text{V}$  (4.8% error).
- **Physical Verification:** Observed power threshold at 0 dBm and second-harmonic phase locking at 9.1 dBm.

**Methodology:** Verified that automated Least-Squares Cross-Validation KDE eliminates the subjectivity and large errors of standard ad-hoc peak detection.

# Outlook and Systematic Challenges

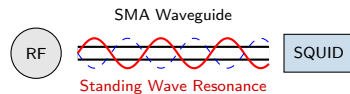
## Impedance Mismatches

Open SMA antenna resonances cause bimodal frequency coupling, yielding 17.3% error in  $\Phi_0$ .

- **PCB Waveguide:** Future designs should employ a matched coplanar waveguide coupler on the SQUID PCB.

## Future Improvements

- **Cryogenic Sensor:** Integrate active temperature sensor to map gap transition  $\Delta(T)$ .
- **Error Propagation:** Propagate LSCV bandwidth errors directly to the linear fit.



SMA cable reflections and power coupling loss



# Acknowledgments

**We sincerely thank:**

**Supervisors & Lab Tutors**

*Oliver Kurtossy, Peter Makk, Szabolcs Csonka, and Andras Halbritter*  
For physical guidance and technical sample support.

**Budapest University of Technology and Economics**  
*Department of Physics & Nanoelectronics Laboratory*

**Project Framework**

*Practice-Oriented Higher Education Infrastructure and Skills Development at BME*  
(Project ID: RRF-2.1.2-21-2022-00005)

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